

Pay attention to the axis, the area under the force displacement gives work done.

9.

Ans: **C**

The distance R from each star to the centre of the circular orbit is $8.0 \times 10^{11} \text{ m}$

The orbital period T is $3.78 \times 10^7 \text{ s}$

Gravitational force provides centripetal force

$$\frac{GMM}{(2R)^2} = MR\omega^2$$

$$M = \frac{16R^3\pi^2}{GT^2}$$

$$M = \frac{16(8 \times 10^{11})^3 \pi^2}{G(3.78 \times 10^7)^2} = 8.5 \times 10^{32} \text{ kg}$$

10.

Ans: **B**